



US012385559B2

(12) **United States Patent**
Chmielewski et al.

(10) **Patent No.:** **US 12,385,559 B2**

(45) **Date of Patent:** **Aug. 12, 2025**

(54) **STRAIN WAVE DRIVE** 7,421,990 B2 * 9/2008 Taye F01L 1/344
123/90.15
(71) Applicant: **Hamilton Sundstrand Corporation,** 8,020,470 B2 9/2011 Saito
Charlotte, NC (US) 10,883,590 B2 * 1/2021 Balsiger H02K 7/116
11,713,108 B2 * 8/2023 Balsiger B64C 9/00
(72) Inventors: **Pawel Chmielewski, Brzezina Łąka** 2015/0354686 A1 * 12/2015 Balsiger H02K 7/116
(PL); **Lukasz Turek, Wrocław (PL);** 74/640
Jonathan Darby, Newcastle (GB); 2016/0305528 A1 * 10/2016 Shahipassand F16H 49/001
Marcin Cis, Lutyńia (PL) 2017/0108107 A1 * 4/2017 Balsiger F16H 49/001
(Continued)

(73) Assignee: **HAMILTON SUNDSTRAND CORPORATION,** Charlotte, NC (US)

FOREIGN PATENT DOCUMENTS

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

CN 109099115 A 12/2018
EP 3165795 A1 5/2017

OTHER PUBLICATIONS

(21) Appl. No.: **18/540,197**

(22) Filed: **Dec. 14, 2023**

(65) **Prior Publication Data**

US 2024/0209929 A1 Jun. 27, 2024

(30) **Foreign Application Priority Data**

Dec. 23, 2022 (EP) 22461653

(51) **Int. Cl.**
F16H 49/00 (2006.01)

(52) **U.S. Cl.**
CPC **F16H 49/001** (2013.01)

(58) **Field of Classification Search**
CPC F16H 49/001; B64C 13/34
USPC 74/640
See application file for complete search history.

(56) **References Cited**

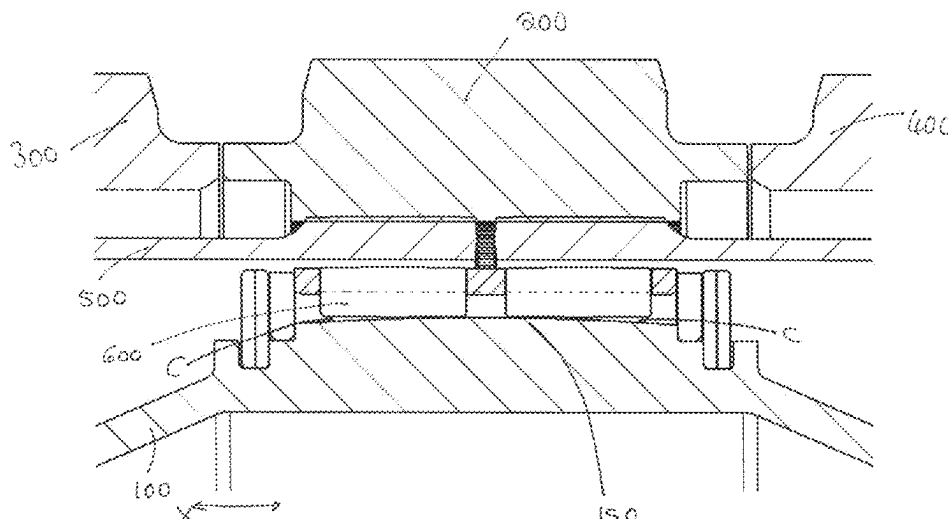
U.S. PATENT DOCUMENTS

3,435,705 A * 4/1969 Musser F16H 49/001
384/127
3,747,434 A 7/1973 Stahlhuth

(57) **ABSTRACT**

A strain wave drive includes an elliptical wave generator shaft rotatable about an axis (X), a flexible flex spline mounted around the wave generator shaft and a ring gear assembly mounted around the flex spline. The ring gear assembly includes an output ring gear having a circular inner periphery and is sandwiched between two earth ring gears each having a circular inner periphery. The flex spline has a first number of radially outwardly extending teeth around its outer periphery and each earth ring gear has the first number of radially inwardly extending teeth to engage with the teeth of the flex spline.

13 Claims, 3 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2019/0011032 A1 * 1/2019 Balsiger F16H 49/001
2019/0107188 A1 * 4/2019 Balsiger G01D 5/04

* cited by examiner

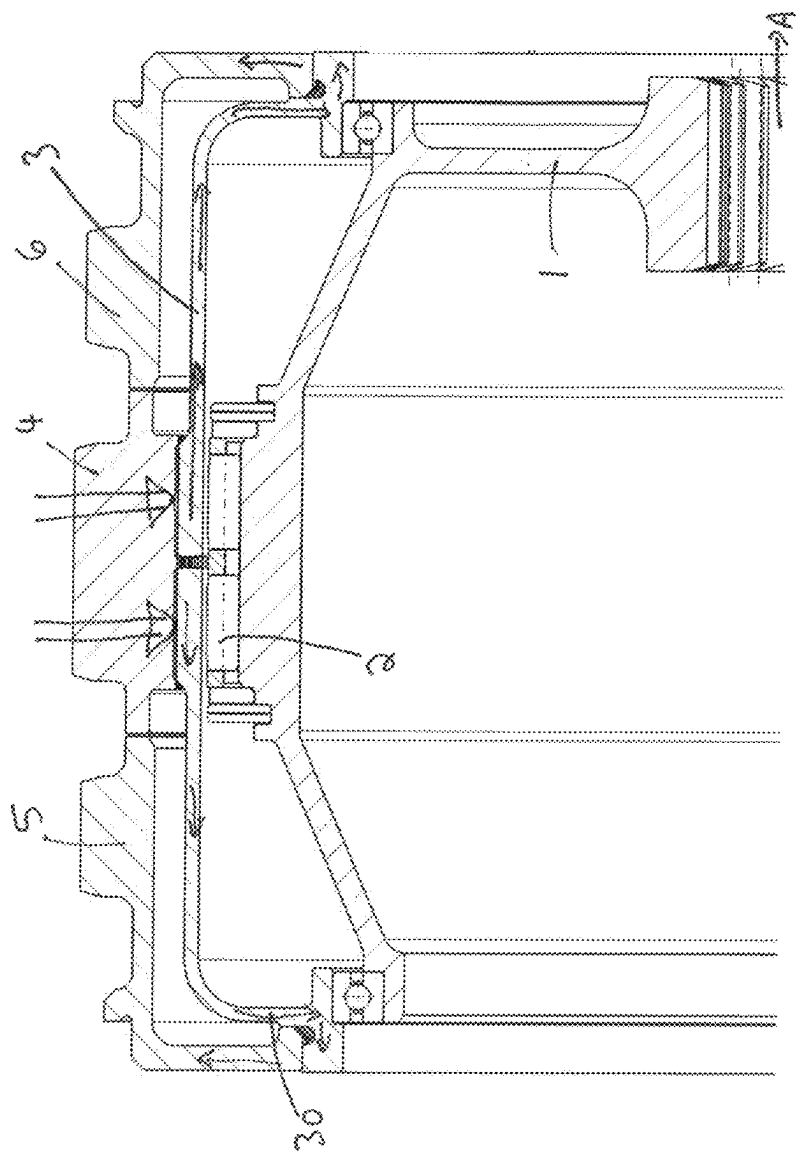


FIG. 1A
PRIOR ART

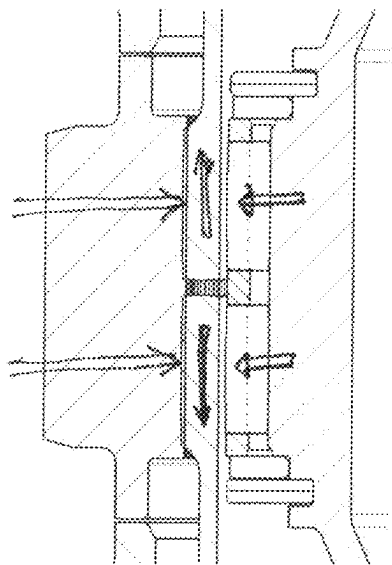


FIG. 1B
PRIOR ART

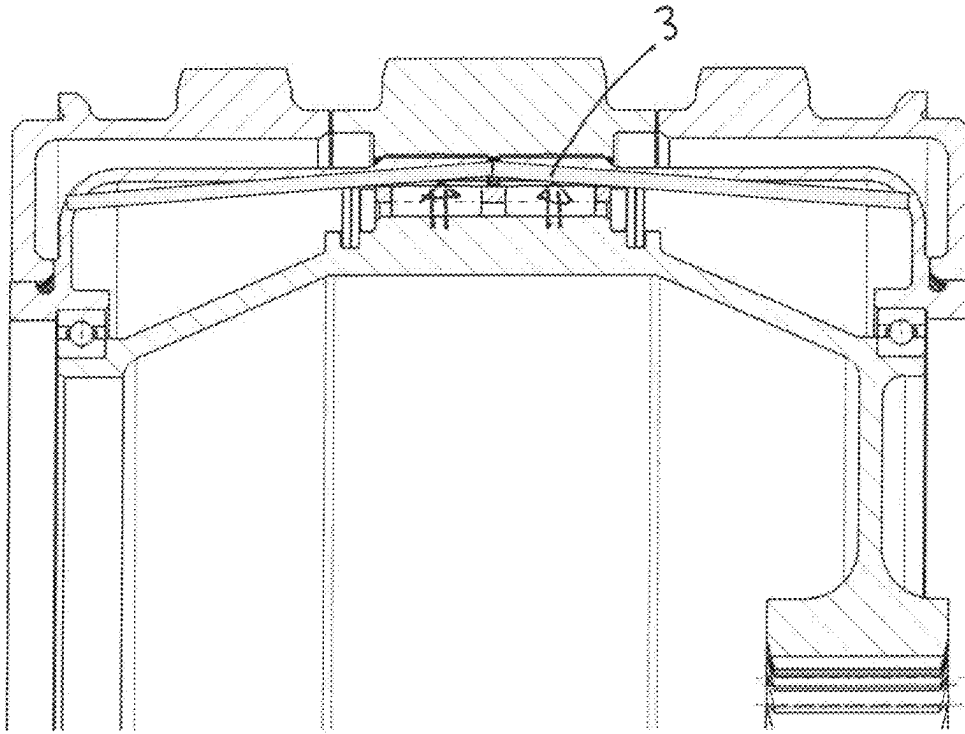


Fig. 2
PRIOR ART

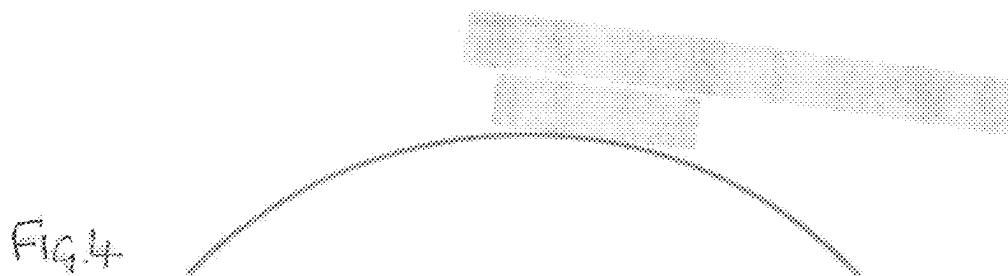
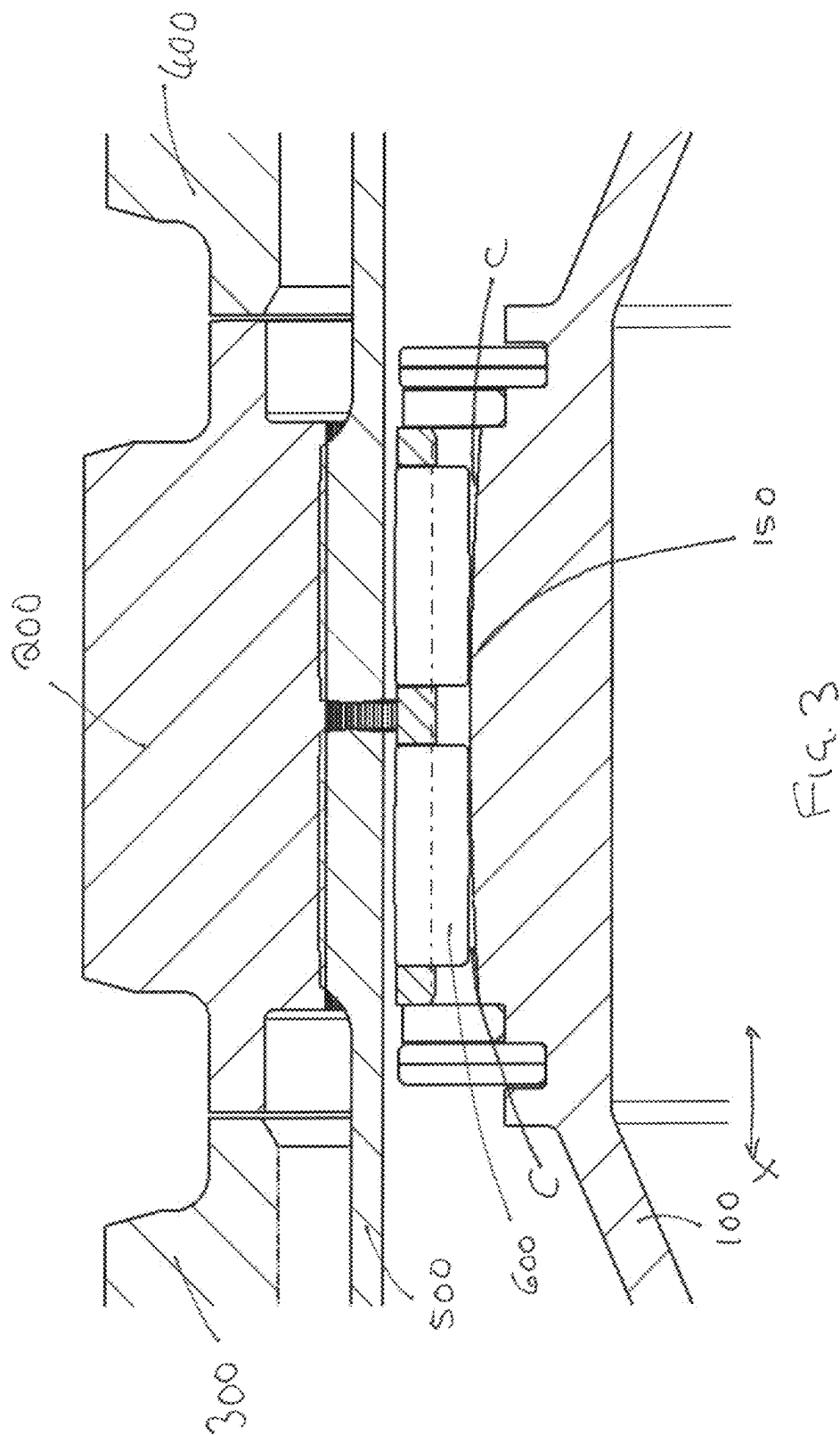


Fig. 4



1

STRAIN WAVE DRIVE**CROSS REFERENCE TO RELATED APPLICATIONS**

This application claims priority to European Patent Application No. 22461653.2 filed Dec. 23, 2022, the entire contents of which is incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a strain wave drive particularly a strain wave drive for rotary actuation of flight control surfaces of an aircraft.

BACKGROUND

Rotary actuation of devices or surfaces e.g. flight control surfaces in an aircraft, requires high torque drive and so typically requires gearing between the electric motor, which is typically high speed, low torque, and the output shaft of the actuator that drives the surface. Conventional involute gears, comprising a number of intermeshing toothed wheels to create the required gear ratio, are relatively large and heavy and the higher the required ratio, the more gears and, thus, the greater the size and weight of the overall actuator system.

In aircraft in particular, and also in other applications, there is a need to minimise the size and weight of components and there is a desire to provide a gear or drive that can provide the required torque ratio using a smaller, lighter, more compact arrangement.

A known type of gear mechanism that is particularly compact is the strain wave drive or gear. Strain wave gearing, also known as a harmonic drive, is used in many drive trains as it is compact and lightweight and can be used to provide a high gear ratio between an input or drive shaft and an output shaft. These properties make strain wave gearing suited to use in aircraft and other vehicles where space is limited and weight of components should be minimised, whilst maintaining reliability and performance.

A strain wave gear system includes a wave generator which is in the form of an elliptical shaft and a compliant ball bearing in which the elliptical shaft rotates. A flexible toothed ring (a flex spline) is mounted about the wave generator and engages, and conforms to the shape of, the output shaft. A fixed outer ring, or circular spline is provided around the flex spline and has inner teeth that engage with the outer teeth of the flex spline but, due to the elliptical shape of the wave generator, the flex spline only engages with the teeth of the outer ring at the major diameter of the wave generator.

In operation, a drive shaft, connected to the wave generator, is rotated e.g. by a motor, which causes rotation of the wave generator. The bearing conforms to the elliptical shape of the wave generator. The flex spline conforms to the shape of the drive shaft and so as the wave generator rotates, the flex spline will only engage with the inner teeth of the outer ring at the major axes of the ellipse. The circular spline has a different number of inner teeth to the number of outer teeth of the flex spline. Rotation of the drive shaft thus causes a slower rotation of the output shaft by its engagement with the flex spline. The output shaft is connected to the device or surface to be moved by the actuator.

Known strain wave drives, however, have to be designed with built-in compliance (as discussed below) to ensure correct load distribution and engagement of the teeth. This

2

adds to the costs and complexity of such gears for use in e.g. flight control rotary actuation. There is, therefore, a desire for an improved strain wave drive that retains the benefits of compactness and light weight whilst overcoming problems associated with existing strain wave drives.

SUMMARY

According to the disclosure, there is provided a strain wave drive comprising: an elliptical wave generator shaft (100) rotatable about an axis (X); a flexible flex spline (500) mounted around the wave generator shaft; a ring gear assembly mounted around the flex spline, the ring gear assembly comprising an output ring gear (200) having a circular inner periphery and being sandwiched between two earth ring gears (300, 400) each having a circular inner periphery; wherein the flex spline has a first number of radially outwardly extending teeth around its outer periphery; and wherein each earth ring gear has the first number of radially inwardly extending teeth to engage with the teeth of the flex spline; and wherein the output ring gear has a second number of radially inwardly extending teeth, such that as the wave generator shaft rotates, it causes the flex spline to take up the elliptical form of the wave generator shaft and the teeth of the flex spline engage with teeth of the output ring gear at the major diameter of the ellipse, causing the output ring gear to rotate; and wherein the wave generator shaft is provided with a radially outer cam surface defining the elliptical shape and arranged to engage with an inner surface of the flex spline; the strain wave drive further comprising roller bearings provided between the cam surface and the flex spline, and wherein the cam surface is a curved surface.

BRIEF DESCRIPTION

Examples of the strain wave drive according to the disclosure will now be described with reference to the drawings. It should be noted that these are examples only and variations are possible within the scope of the claims.

FIG. 1A shows a cross-section through a known type of strain wave drive.

FIG. 1B shows a detail of the strain wave drive of FIG. 1A.

FIG. 2 shows a strain wave drive such as shown in FIGS. 1A and 1B during deflection of the flex spline.

FIG. 3 shows an example of a strain wave drive in accordance with the present disclosure.

FIG. 4 shows, schematically the relationship between the bearings and the wave generator in a strain wave drive such as shown in FIG. 3.

DETAILED DESCRIPTION

As mentioned above, conventional drives for moving e.g. flight control surfaces in aircraft, have used a series of inter-meshing gear wheels. In an attempt to reduce the overall size and weight of the gearing, thought has been given to the use of the more compact strain wave gears or drives (also known as harmonic drives). Such strain wave drives essentially consist of three main parts: a wave generator, driven by the motor drive shaft, is an elliptical shaft having bearings arranged around the outer perimeter thereof. The wave generator is located within a cylindrical flex spline which is a flexible annular component having radially outwardly extending teeth. The flex spline is sufficiently flexible to take up the elliptical shape of the wave generator as the wave generator and its bearings rotate within the flex

3

spline. A rigid circular spline is a ring that fits around the flex spline. The circular spline has inwardly extending teeth. The circular spline is typically fixed e.g. to the actuator housing. As the wave generator is rotated by the motor, it causes the flex spline to take up the elliptical shape such that the outwardly extending teeth of the flex spline mesh with the inwardly extending teeth of the circular spline at the locations of the major axis of the ellipse. The circular spline typically has more teeth than the flex spline such that as the teeth engage, the flex spline is caused to rotate relative to the circular spline at a rate of rotation different to that of the motor. In other applications, such as shown in FIG. 1A, an output ring gear is positioned around the flex spline. The output ring gear is provided with inner teeth that engage with the teeth of the flex spline. As the flex spline deforms due to rotation of the wave generator inside it, its teeth engage with the inwardly protruding teeth of the output ring and, due to the elliptical movement of the point of engagement, this causes the output ring gear to rotate according to the gear ratio. The output ring gear is connected with a part or component to be moved by the drive.

Current designs of strain wave drives for rotary actuation of flight control surfaces require a right-angled U-shaped or cup-shaped flex spline to provide the flex spline function within the available space envelope and with the required strength and stress resistance. The right-angle cup shape adds stiffness and distributes stress through the earthed parts of the system. An example of such a strain wave drive is shown in FIGS. 1A and 1B. Because the flex spline is cup-shaped, it has to be formed as two cup parts for manufacturing and assembly purposes, the two cup parts meeting at a join aligned with the output ring gear sandwiched between the earth gears.

FIG. 1A shows a section of the strain wave drive, showing the wave generator 1, rotatable about axis A. Needle bearings 2 are provided around the perimeter of the wave generator 1. The flex spline 3 is fitted around the wave generator to engage with the bearings to take up the elliptical shape of the wave generator. In the example shown, two flex splines are provided, each formed with a flange 30. Teeth (not shown in this figure) on the outer surface of the flex spline engage, at certain positions of rotation, with inwardly extending teeth (not shown in this figure) of the output ring gear 4. The earth gears may be fixed relative to the actuator housing or other fixed component and the flex spline may engage with an output shaft or output ring gear 4 which is caused to rotate with rotation of the point of engagement with the flex spline as the wave generator rotates. The output gear rotates according to the gear ratio and may be fixed to a part or surface to be rotated e.g. a flight control surface.

Because there is uneven loading on the flex spline 3 between the wave generator, via the bearings, and the output gear, and equal loads are transmitted from the flex spline through the earth gears, as the flex spline deforms to the shape of the bearings, it creates a 'coning' effect as shown in FIG. 2. The split, two-part flex spline forms a conical shape at its interface with the output gear and the earth gears which can cause an uneven interference between the intermeshing teeth at some positions, i.e. when the flex spline deflects, it engages with the corresponding teeth at an angle. This angle may also vary under changing loads and temperatures.

To take into account the tilt between the meshing gears when the flex spline is deformed, compliance needs to be built into the design to ensure uniform load distribution from the parts that generate the motion of the gearing through to the flex spline itself. This has conventionally been done by

4

adding additional components to provide the required compliance. These components, however, add to the cost, complexity, size and weight of the drive.

To address these problems, the strain wave drive of this disclosure incorporates compliance into the drive by modifying the shape of the outer cam surface of the wave generator at the interface with the bearings. This can be best seen in FIG. 3 where C represents the curved profile of the wave generator cam relative to the straight bearings. FIG. 4 shows this in an exaggerated view of explanation. The cam surface is shaped in a barrel shape as opposed to providing a straight or flat surface as is conventionally the case. The compliance in the shaping of the wave generator cam surface allows the use of straight bearings (needle bearings). The bearings will self-align—i.e. find their own natural operating angle—when driven by the wave generator cam profile and depending on the length of the flex spline, which determines the operating angle.

The compliance built into the wave generator and through the self-alignment of the rollers means that the rollers will make contact with the flex spline such that uniform loading will be transferred through to the inner surface of the flex spline. This uniform loading ensures that the coning effect is compensated.

The elliptical wave generator 100 is a shaft, rotatable about an axis X, and located within the flex spline 500 which is a cup-shaped component. Mounted around the flex spline is the output ring gear 200 between the earth gears 300, 400. The flex spline 500 takes the elliptical shape of the wave generator and therefore engages with the teeth on the output gear 200 at the points of the ellipse major axis. Because the output gear has two more teeth than the flex spline, there is relative movement in the rotational direction between the flex spline and the output gear.

The output ring gear 200 has two teeth more than the flex spline 500 and this provides the gear ratio.

The bearings, here needle rollers 600, are fitted around the wave generator 100 on a cam 150 thereof. The bearings 600 rotate with the wave generator 100 and impart the elliptical shape of the wave generator to the flex spline 500. As the elliptical wave generator rotates, the outer sections of the flex spline are secured by the earth gears and, at the major diameter of the ellipse, the middle section of the flex spline is deformed to take up the elliptical shape and its teeth, in that section, mesh with the teeth of the output gear causing it to rotate.

The surface C of the cam onto which the bearings 600 are mounted is, as mentioned above, formed with a curved or barrel profile (see FIG. 4) to provide the required compliance and which allows the rollers to self-align.

Because of the shape of the wave generator, the strain wave drive has a compliant, self-aligning design that provides uniform stress distribution loading. The cam profile is precision machined and allows the use of off-the-shelf roller bearings. As the bearings themselves do not need to be specially shaped or formed to provide compliance, there is a reduction in tolerance variation and the cost and complexity of manufacture and assembly is reduced.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not

5

preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims

The invention claimed is:

1. A strain wave drive comprising:

an elliptical wave generator shaft rotatable about an axis (X);

a flexible flex spline mounted around the wave generator shaft; and

a ring gear assembly mounted around the flex spline, the ring gear assembly comprising an output ring gear having a circular inner periphery and being sandwiched between two earth ring gears each having a circular inner periphery;

wherein the flex spline has a first number of radially outwardly extending teeth around its outer periphery; and

wherein each earth ring gear has the first number of radially inwardly extending teeth to engage with the teeth of the flex spline;

wherein the output ring gear has a second number of radially inwardly extending teeth, such that as the elliptical wave generator shaft rotates, the elliptical wave generator shaft causes the flex spline to take up an elliptical form of the elliptical wave generator shaft and the teeth of the flex spline engage with the teeth of the output ring gear at a major diameter of the elliptical form, causing the output ring gear to rotate; and

6

wherein the elliptical wave generator shaft is provided with a radially outer cam surface defining the elliptical form and arranged to engage with an inner surface of the flex spline; and

the strain wave drive further comprising:

roller bearings provided between the cam surface and the flex spline, and wherein the cam surface is a curved surface;

wherein the flex spline is a tubular flex spline mounted around the wave generator shaft and has flanges at opposing ends.

2. The strain wave drive of claim 1, wherein the roller bearings have a straight surface that contacts the curved cam surface.

3. The strain wave of claim 1, wherein the roller bearings are needle roller bearings.

4. The strain wave drive of claim 1, wherein the second number is greater than the first number.

5. The strain wave drive of claim 4, wherein the second number is two more than the first number.

6. The strain wave drive of claim 4, wherein the second number is three more than the first number.

7. The strain wave drive of claim 1, wherein the earth ring gears are configured to be attached to an external fixed part.

8. The strain wave drive of claim 7, wherein the earth ring gears are provided with attachment flanges for attachment to the external fixed part.

9. The strain wave drive of claim 1, wherein the output ring gear is configured to be attached to a part or component to be moved.

10. The strain wave drive of claim 9, wherein the output ring gear is provided with a flange for attachment to the part or component to be moved.

11. The strain wave drive of claim 1, wherein the flex spline is formed as two cup-shaped parts adjoined at their open ends.

12. The strain wave drive of claim 1, for moving a moveable part or component of an aircraft.

13. The strain wave drive of claim 1, for moving a flight control surface of an aircraft.

* * * * *